

time series modelling project draft writeup

January 6, 2026

1 formulating the multi-factor model

- why chose the model (comes from grinold and kahn, widely known)
- give the form of the model

1.1 producing forecasts

-discuss MSCI barra flow

We largely follow the procedure described in the first three chapters of the Barra Risk Model Handbook (Barra, 2004). This involves selecting and standardizing descriptors to compute the factor exposures of the securities. Once the exposure matrix is constructed,

1.2 choosing factors

An obvious place to start is the Fama-French three-factor model. The issue with this is the SMB and HML factors require data about firms that isn't freely available, which makes them unsuitable for a small proof-of-concept implementation like ours. In the absence of fundamental firm data, we seek to supplement the market exposure factor with factors computed based only on daily price and volume data freely available on the internet. These additional factors should be intuitively motivated and widely documented in the literature.

We make the following definitions for security i at time t , whose adjusted close price we denote by $p_{i,t}$. The number of shares traded on day t is $v_{i,t}$. The adjusted close price of the market index at this time is $p_{m,t}$. We define the momentum descriptor by

$$d_{i,t} := \frac{p_{i,t-21}}{p_{i,t-252}} - 1, \quad (1.2.1)$$

following the “eleven-month returns lagged one month” definition given on page 61 of (Carhart, 1997), which itself cites (Jegadeesh & Titman, 1993). Since we're using daily data, we include the past 252 and exclude the past 21 trading days. This is done to remove the effect of short-term mean reversion, which we deal with separately. Adding this factor to the Fama-French three-factor model results in the Carhart four-factor model.

We define the market exposure descriptor by

$$d_{i,t} := \hat{\beta}_{i,t}, \quad (1.2.2)$$

where $\hat{\beta}_{i,t}$ is an estimate of the market exposure β_i from the single-index model

$$r_i = \alpha_i + \beta_i r_m + \varepsilon_i, \quad (1.2.3)$$

where r_i and r_m are the returns to security i and the market index over some period. We compute $\hat{\beta}_{i,t}$ based on historical daily returns

$$r_{i,\tau} := \frac{p_{i,\tau} - p_{i,\tau-1}}{p_{i,\tau-1}}, \quad r_{m,\tau} := \frac{p_{m,\tau} - p_{m,\tau-1}}{p_{m,\tau-1}}, \quad (1.2.4)$$

for τ in a lookback window $[t - w + 1, t]$ of length w . Chapter 7 of (Elton, Gruber, Brown, & Goetzmann, 2014) estimates $\hat{\beta}_{i,t}$ using ordinary least-squares regression, but other methods are possible. Rather than doing explicit OLS regression, we compute

$$\beta_i = \frac{\text{Cov}(r_i, r_m)}{\text{Var}(r_m)} \quad (1.2.5)$$

using sample covariance and variance over the lookback window, which should be equivalent (verify this!!).

We define the idiosyncratic volatility descriptor by

$$d_{i,t} := \sqrt{\frac{1}{w-1} \sum_{\tau \in [t-w+1, t]} r_{i,\tau} - \hat{\beta}_{i,t} r_{m,\tau}}, \quad (1.2.6)$$

which is simply the sample standard deviation of the realized idiosyncratic return on the security during the lookback window, given our estimate of the market exposure. Extensive discussion of idiosyncratic volatility and average return is given in (Ang, Hodrick, Xing, & Zhang, 2006), which defines it on page 18.

We define the total volatility descriptor by

$$d_{i,t} := \sqrt{\frac{1}{w-1} \sum_{\tau \in [t-w+1, t]} r_{i,\tau} - \bar{r}_i}, \quad (1.2.7)$$

where $\bar{r}_i$ is the mean daily return during the lookback window. This descriptor is the sample standard deviation of the realized daily return on the security during the lookback window. This is also discussed in (Ang et al., 2006).

We define the short-term reversal descriptor by

$$d_{i,t} := -\frac{p_{i,t}}{p_{i,t-21}} + 1. \quad (1.2.8)$$

This is the additive inverse of the cumulative return on the security over the past month. The significance of short-term reversal over one month is discussed in (Da, Liu, & Schaumburg, 2014).

We define the illiquidity descriptor by

$$d_{i,t} := \frac{1}{252} \sum_{\tau=t-251}^t \frac{|r_{i,\tau}|}{v_{i,\tau} p_{i,\tau}}, \quad (1.2.9)$$

following the definition given on page 34 of (Amihud, 2002). We compute the daily volume in dollars as the product of the number of shares and the adjusted close price.

1.3 forming factor exposure matrix X

-normalization (demeaning)

- 1.4 forming factor returns f
- 1.5 forming factor covariance matrix F
- 1.6 forming individual risks D
- 1.7 forecasting
- 1.8 formulating the portfolio optimization problem

2 implementation

- details of implementation
- use pandas for time series data and fast rolling calculations
- perhaps format of dataframes?
- back fill missing data
- data cleaning stuff? excluding vs including securities
- yfinance related stuff

3 results

- numerical results with benchmark and no benchmark
- interpret plot checking and idiot checks
- justify removing illiq
- performance metrics and plots

References

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